

VERTICAL LINE TEST. A curve in the xy -plane is a function if and only if every vertical line intersects the curve at most once.

◆ **EXAMPLE 2** Using the vertical line test, explain why each curve in Figure 5 does or does not define a function.

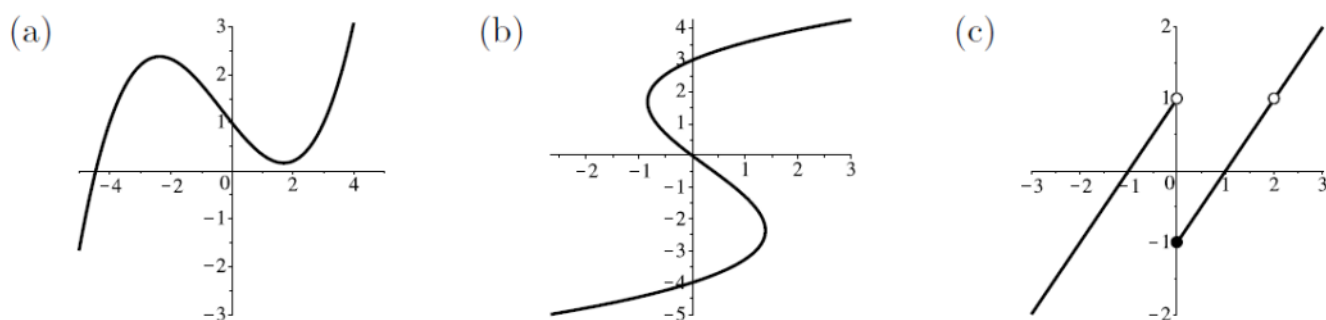


Figure 5: Curves for Example 2

► **QUESTION 2** Using the vertical line test, explain why each curve in Figure 7 does or does not define a function.

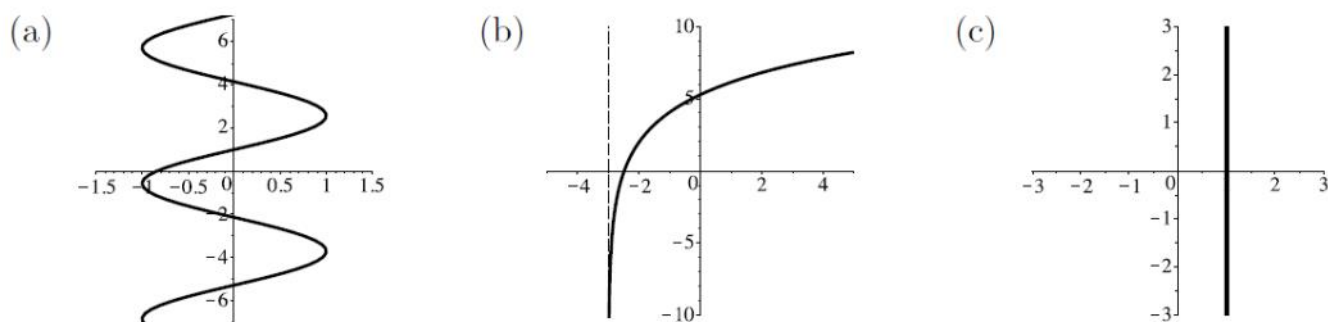


Figure 7: Curves for Question 2

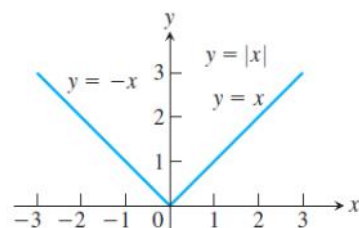
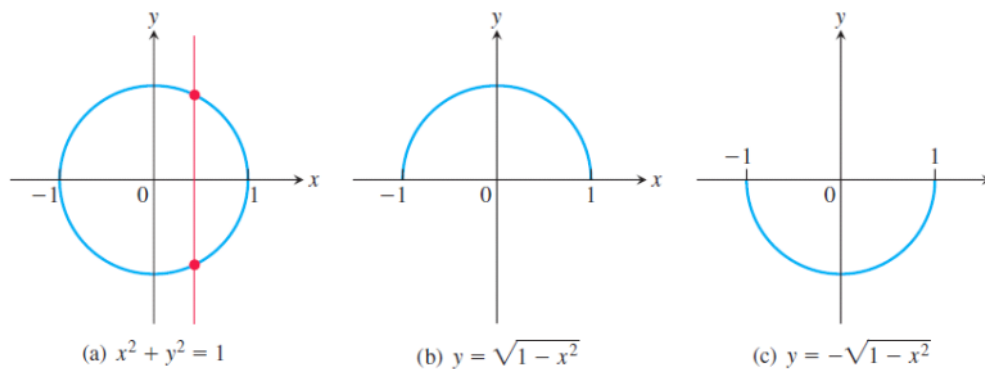


FIGURE 1.8 The absolute value function has domain $(-\infty, \infty)$ and range $[0, \infty)$.



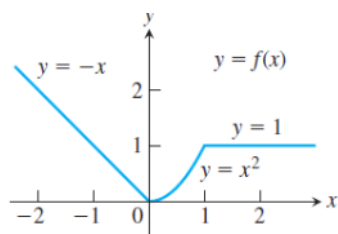
Piecewise-Defined Functions

Sometimes a function is described in pieces by using different formulas on different parts of its domain. One example is the **absolute value function**

$$|x| = \begin{cases} x, & x \geq 0 \\ -x, & x < 0, \end{cases}$$

First formula
Second formula

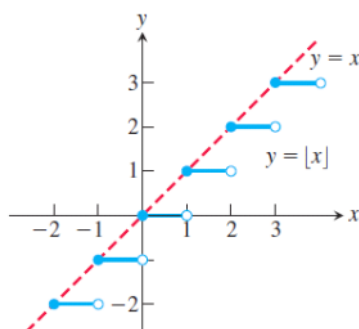
whose graph is given in Figure 1.8. The right-hand side of the equation means that the function equals x if $x \geq 0$, and equals $-x$ if $x < 0$. Piecewise-defined functions often arise when real-world data are modeled. Here are some other examples.



EXAMPLE 4 The function

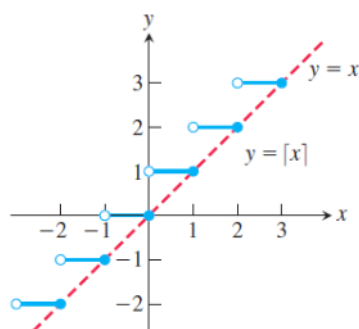
$$f(x) = \begin{cases} -x, & x < 0 & \text{First formula} \\ x^2, & 0 \leq x \leq 1 & \text{Second formula} \\ 1, & x > 1 & \text{Third formula} \end{cases}$$

is defined on the entire real line but has values given by different formulas, depending on the position of x . The values of f are given by $y = -x$ when $x < 0$, $y = x^2$ when $0 \leq x \leq 1$, and $y = 1$ when $x > 1$. The function, however, is *just one function* whose domain is the entire set of real numbers (Figure 1.9). ■



EXAMPLE 5 The function whose value at any number x is the *greatest integer less than or equal to* x is called the **greatest integer function** or the **integer floor function**. It is denoted $\lfloor x \rfloor$. Figure 1.10 shows the graph. Observe that

$$\begin{aligned} \lfloor 2.4 \rfloor &= 2, & \lfloor 1.9 \rfloor &= 1, & \lfloor 0 \rfloor &= 0, & \lfloor -1.2 \rfloor &= -2, \\ \lfloor 2 \rfloor &= 2, & \lfloor 0.2 \rfloor &= 0, & \lfloor -0.3 \rfloor &= -1, & \lfloor -2 \rfloor &= -2. \end{aligned}$$



EXAMPLE 6 The function whose value at any number x is the *smallest integer greater than or equal to* x is called the **least integer function** or the **integer ceiling function**. It is denoted $\lceil x \rceil$. Figure 1.11 shows the graph. For positive values of x , this function might represent, for example, the cost of parking x hours in a parking lot that charges \$1 for each hour or part of an hour. ■

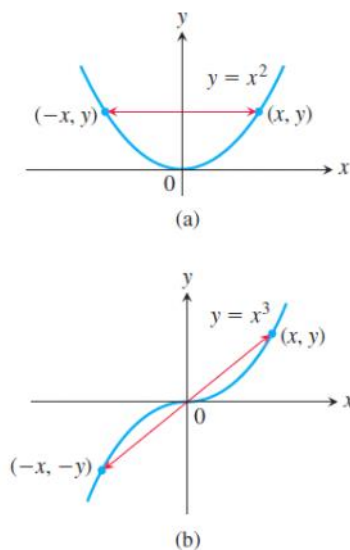


FIGURE 1.12 (a) The graph of $y = x^2$ (an even function) is symmetric about the y -axis. (b) The graph of $y = x^3$ (an odd function) is symmetric about the origin.

EXAMPLE 8 Here are several functions illustrating the definition.

$f(x) = x^2$	Even function: $(-x)^2 = x^2$ for all x ; symmetry about y -axis.
$f(x) = x^2 + 1$	Even function: $(-x)^2 + 1 = x^2 + 1$ for all x ; symmetry about y -axis (Figure 1.13a).
$f(x) = x$	Odd function: $(-x) = -x$ for all x ; symmetry about the origin.
$f(x) = x + 1$	Not odd: $f(-x) = -x + 1$, but $-f(x) = -x - 1$. The two are not equal. Not even: $(-x) + 1 \neq x + 1$ for all $x \neq 0$ (Figure 1.13b). ■

Sums, Differences, Products, and Quotients

Like numbers, functions can be added, subtracted, multiplied, and divided (except where the denominator is zero) to produce new functions. If f and g are functions, then for every x that belongs to the domains of both f and g (that is, for $x \in D(f) \cap D(g)$), we define functions $f + g$, $f - g$, and fg by the formulas

$$\begin{aligned}(f + g)(x) &= f(x) + g(x) \\ (f - g)(x) &= f(x) - g(x) \\ (fg)(x) &= f(x)g(x).\end{aligned}$$

Notice that the $+$ sign on the left-hand side of the first equation represents the operation of addition of *functions*, whereas the $+$ on the right-hand side of the equation means addition of the real numbers $f(x)$ and $g(x)$.

At any point of $D(f) \cap D(g)$ at which $g(x) \neq 0$, we can also define the function f/g by the formula

$$\left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)} \quad (\text{where } g(x) \neq 0).$$

Functions can also be multiplied by constants: If c is a real number, then the function cf is defined for all x in the domain of f by

$$(cf)(x) = cf(x).$$

EXAMPLE 1 The functions defined by the formulas

$$f(x) = \sqrt{x} \quad \text{and} \quad g(x) = \sqrt{1-x}$$

have domains $D(f) = [0, \infty)$ and $D(g) = (-\infty, 1]$. The points common to these domains are the points

$$[0, \infty) \cap (-\infty, 1] = [0, 1].$$

The following table summarizes the formulas and domains for the various algebraic combinations of the two functions. We also write $f \cdot g$ for the product function fg .

Function	Formula	Domain
$f + g$	$(f + g)(x) = \sqrt{x} + \sqrt{1-x}$	$[0, 1] = D(f) \cap D(g)$
$f - g$	$(f - g)(x) = \sqrt{x} - \sqrt{1-x}$	$[0, 1]$
$g - f$	$(g - f)(x) = \sqrt{1-x} - \sqrt{x}$	$[0, 1]$
$f \cdot g$	$(f \cdot g)(x) = f(x)g(x) = \sqrt{x(1-x)}$	$[0, 1]$
f/g	$\frac{f}{g}(x) = \frac{f(x)}{g(x)} = \sqrt{\frac{x}{1-x}}$	$[0, 1)(x = 1 \text{ excluded})$
g/f	$\frac{g}{f}(x) = \frac{g(x)}{f(x)} = \sqrt{\frac{1-x}{x}}$	$(0, 1](x = 0 \text{ excluded})$



Transformations

Transformations of Functions:



In Summary...

Special Case: $A = -1$
Vertical Reflection

Vertical Scaling

Vertical Shift

$$y = Af(B(x + C)) + D$$

Special Case: $B = -1$
Horizontal Reflection

Horizontal Scaling

Horizontal Shift

Shift Up
& Down

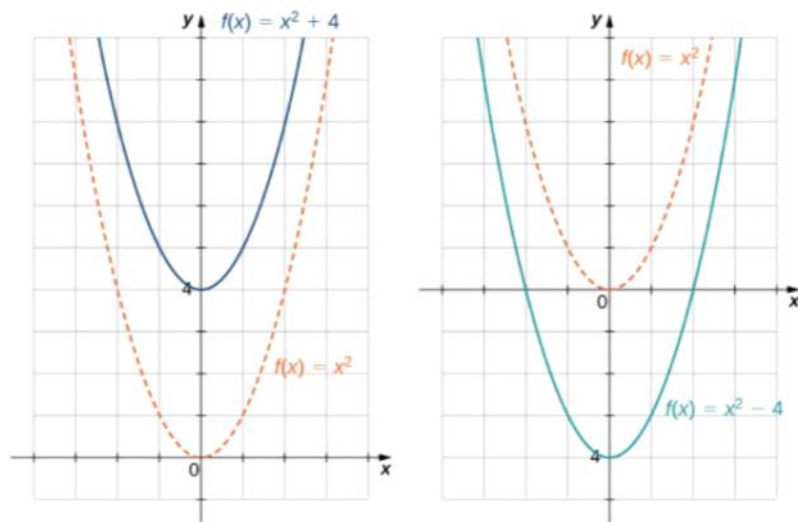


Vertical Shift

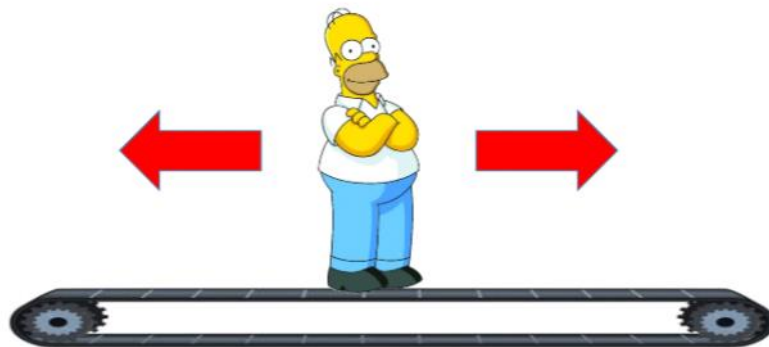
$$y = f(x) + D$$

$D > 0$: shift up

$D < 0$: shift down



Shift Left
& Right

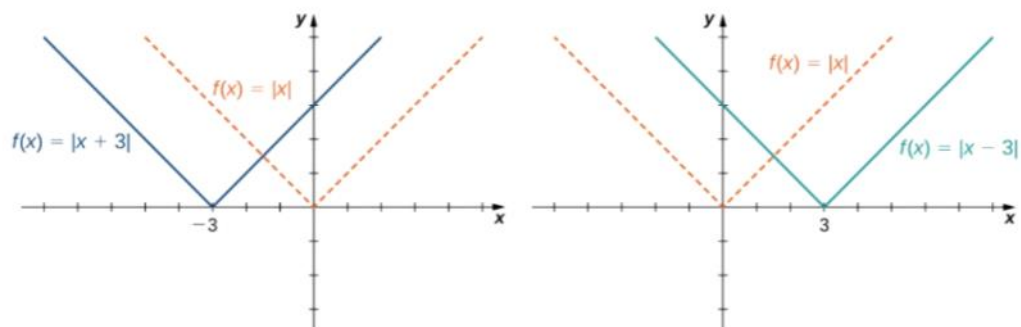


Horizontal Shift

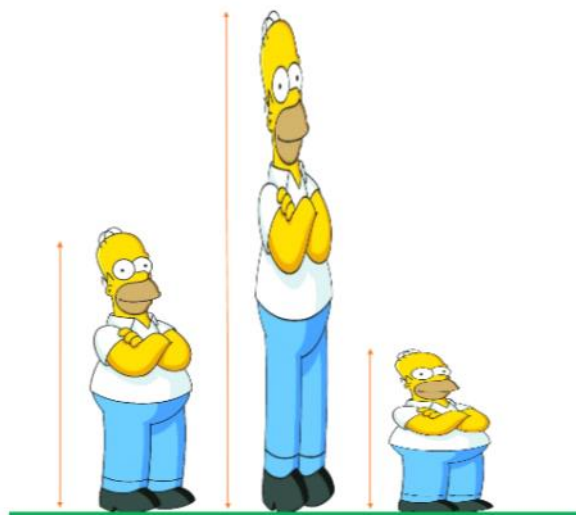
$$y = f(x + C)$$

$C > 0$: shift left

$C < 0$: shift right



Vertical Stretch & Compress

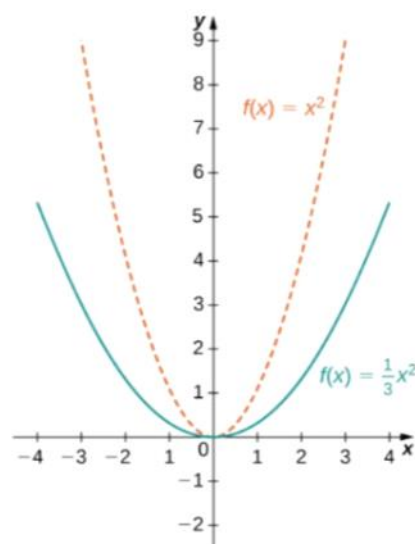
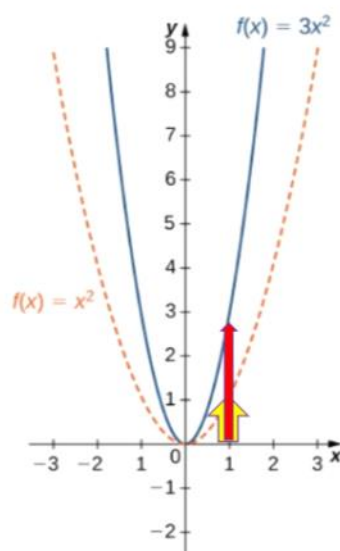


Vertical Scaling

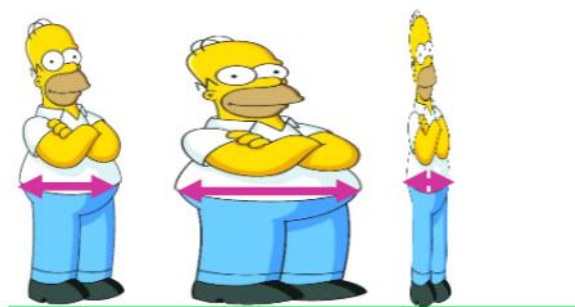
$$y = Af(x)$$

$A > 1$: Vertical Stretch

$0 < A < 1$: Vertical Compress



Horizontal Stretch & Compress

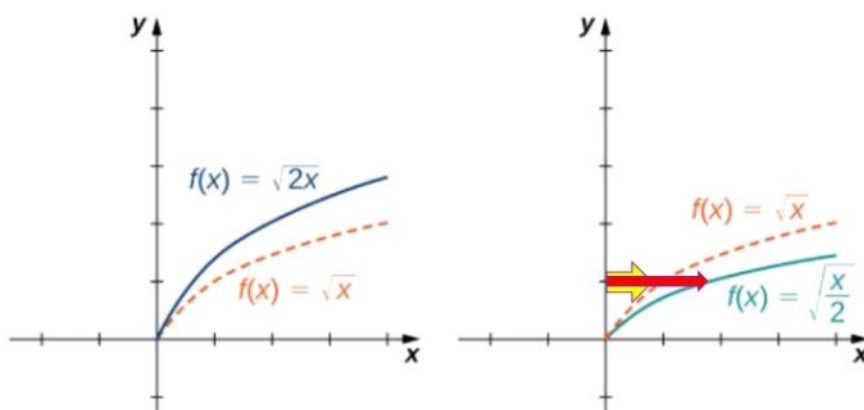


Horizontal Scaling

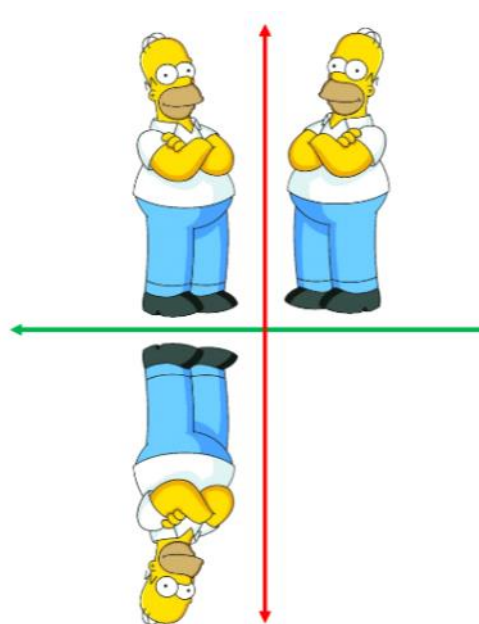
$$y = f(Bx)$$

$B > 1$: Horizontal Compress

$0 < B < 1$: Horizontal Stretch



Flip Horizontal
& Vertical



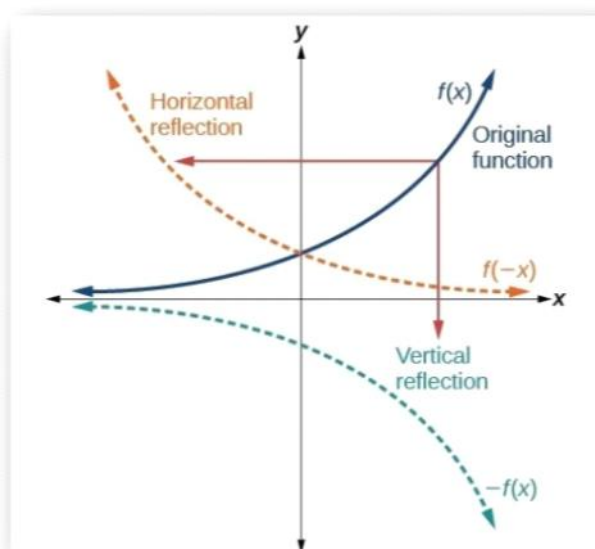
Special Case:
Reflection

$$y = f(Bx) \quad B = -1$$

Horizontal Reflection

$$y = Af(x) \quad A = -1$$

Vertical Reflection



This is what your mobile calls “flip vertical” and “flip horizontal”

In Summary...

Special Case: $A = -1$
Vertical Reflection

Vertical Scaling

Vertical Shift

$$y = Af(B(x + C)) + D$$

Special Case: $B = -1$
Horizontal Reflection

Horizontal Scaling

Horizontal Shift

Or, we could try a quick
reference value check

Let

$$f(0) = R$$

$$f(1) = S$$

Reference

Transformed

$$f(x) = R, \\ \text{at } x = 0$$

$$f(x + 3) = R, \\ \text{at } x = -3$$

R has shifted left
by 3 units

$$f(x) = R, \\ \text{at } x = 0$$

$$f(x) + 3 = R + 3, \\ \text{at } x = 0$$

R has shifted up by
3 units

$$f(x) = R, \\ \text{at } x = 0$$

$$3f(x) = 3R, \\ \text{at } x = 0$$

R has been scaled
by a factor of 3

$$f(x) = S, \\ \text{at } x = 1$$

$$f(3x) = S, \\ \text{at } x = 1/3$$

S occurs 3 times
earlier

$f(x) + a$ $f(x)$ is shifted upwards by a units

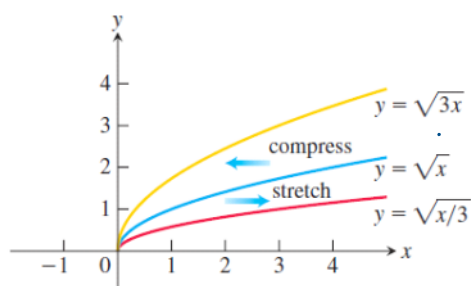
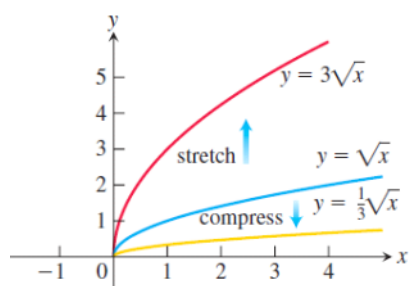
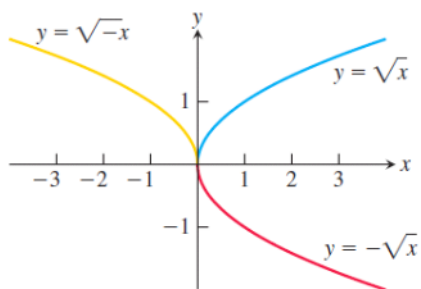
$f(x - a)$ $f(x)$ is shifted rightwards by a units

$-f(x)$ $f(x)$ is reflected by the X-Axis

$f(-x)$ $f(x)$ is reflected by the Y-Axis

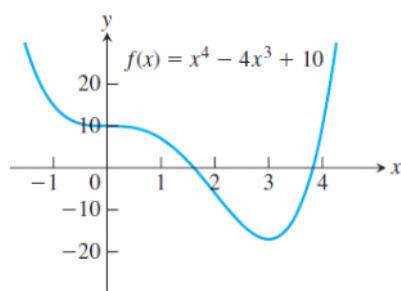
$kf(x)$ $f(x)$ is stretched vertically (or parallel to the Y-Axis) by factor of k

$f(kx)$ $f(x)$ is compressed horizontally (or parallel to the X-Axis) by factor of k

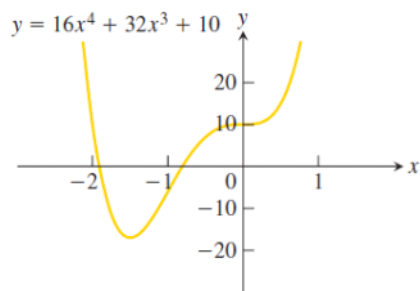


EXAMPLE 5 Given the function $f(x) = x^4 - 4x^3 + 10$ (Figure 1.35a), find formulas to

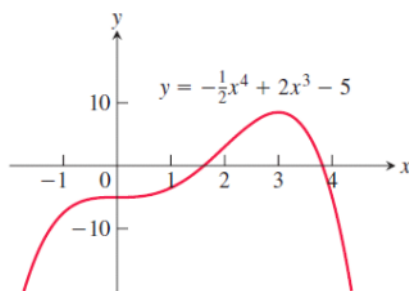
- compress the graph horizontally by a factor of 2 followed by a reflection across the y -axis (Figure 1.35b).
- compress the graph vertically by a factor of 2 followed by a reflection across the x -axis (Figure 1.35c).



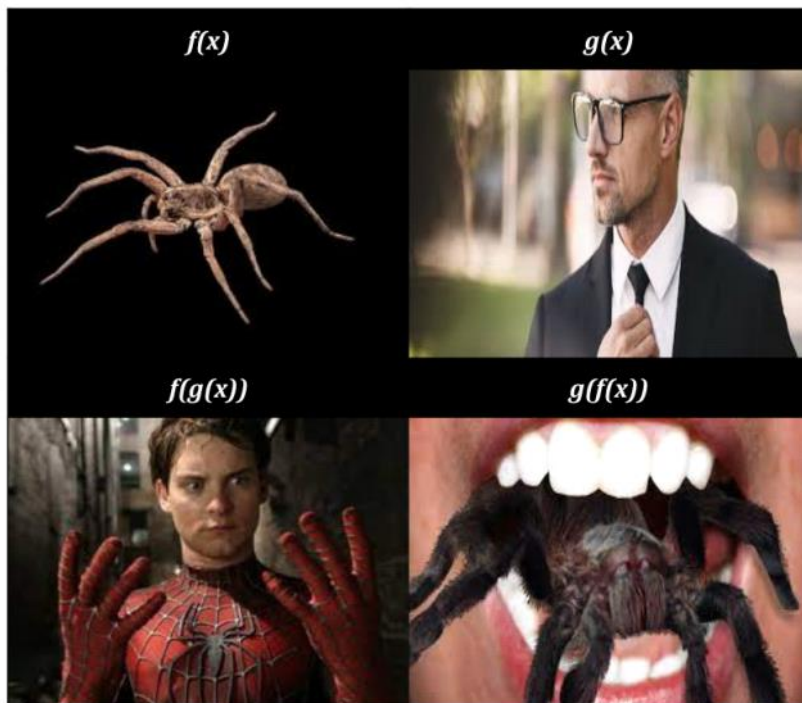
(a)



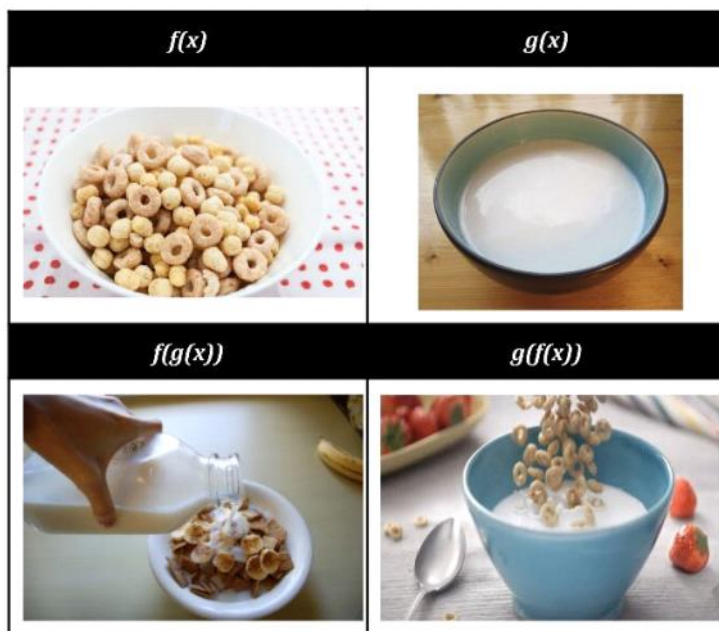
(b)



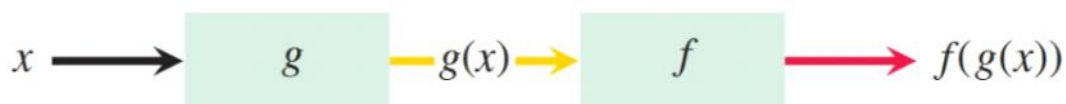
(c)



Meme Artist : Wyatt Fielder



Function Composition



Notation

$$(f \circ g)(x) = f(g(x)).$$

Q. Is function composition commutative?

In general, NO!

$$f(g(x)) \neq g(f(x))$$

E.g., try these

$$f(x) = \sqrt{x} \qquad g(x) = x + 1$$

$$(f \circ g)(x) = f(g(x)) =$$

$$(g \circ f)(x) = g(f(x)) =$$

EXAMPLE 2 If $f(x) = \sqrt{x}$ and $g(x) = x + 1$, find

- (a) $(f \circ g)(x)$ (b) $(g \circ f)(x)$ (c) $(f \circ f)(x)$ (d) $(g \circ g)(x)$.

Solution

Composite	Domain
(a) $(f \circ g)(x) = f(g(x)) = \sqrt{g(x)} = \sqrt{x+1}$	$[-1, \infty)$
(b) $(g \circ f)(x) = g(f(x)) = f(x) + 1 = \sqrt{x} + 1$	$[0, \infty)$
(c) $(f \circ f)(x) = f(f(x)) = \sqrt{f(x)} = \sqrt{\sqrt{x}} = x^{1/4}$	$[0, \infty)$
(d) $(g \circ g)(x) = g(g(x)) = g(x) + 1 = (x+1) + 1 = x+2$	$(-\infty, \infty)$

To see why the domain of $f \circ g$ is $[-1, \infty)$, notice that $g(x) = x + 1$ is defined for all real x but belongs to the domain of f only if $x + 1 \geq 0$, that is to say, when $x \geq -1$. ■

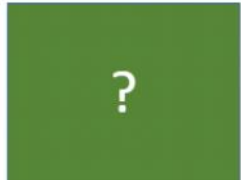
Notice that if $f(x) = x^2$ and $g(x) = \sqrt{x}$, then $(f \circ g)(x) = (\sqrt{x})^2 = x$. However, the domain of $f \circ g$ is $[0, \infty)$, not $(-\infty, \infty)$, since $\sqrt{x}$ requires $x \geq 0$.

A more convincing proof...

Function Composition
is NOT Commutative!

$f(x) =$  $g(x) =$ 

$f(g(x)) =$ 

$g(f(x)) =$ 

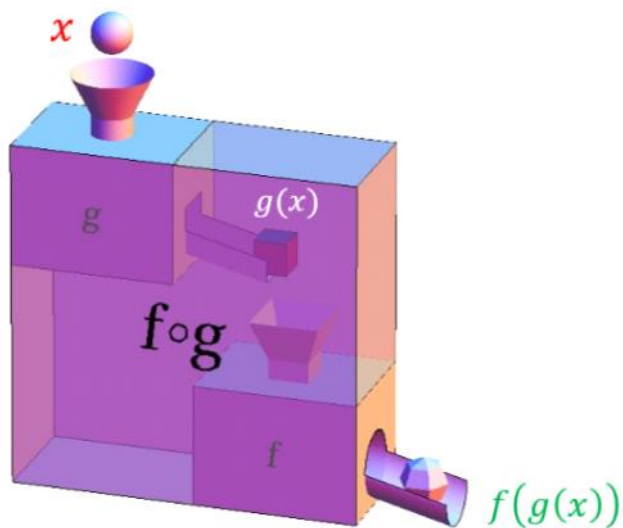
A more convincing proof...

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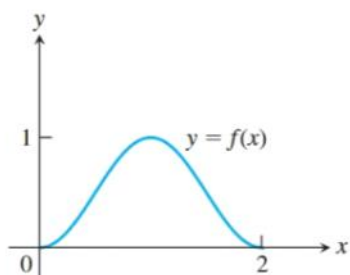
$f(g(x)) =$ 

$g(f(x)) =$ 



Example Problem 3

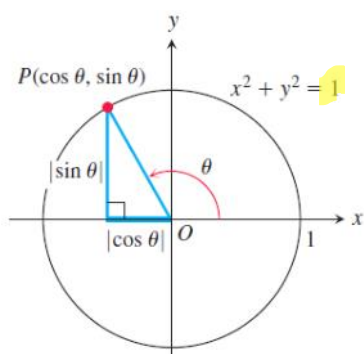
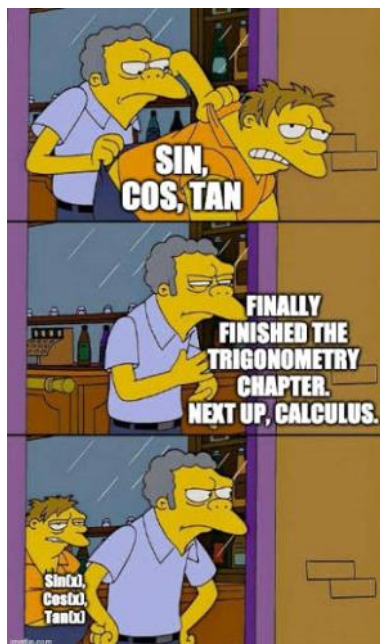
Given



Find

- | | |
|---------------|--------------------|
| a. $f(x) + 2$ | b. $f(x) - 1$ |
| c. $2f(x)$ | d. $-f(x)$ |
| e. $f(x + 2)$ | f. $f(x - 1)$ |
| g. $f(-x)$ | h. $-f(x + 1) + 1$ |

Also try $f(2x)$



$$x = r \cos \theta, \quad y = r \sin \theta.$$

$$\cos^2 \theta + \sin^2 \theta = 1.$$

$$1 + \tan^2 \theta = \sec^2 \theta$$

$$1 + \cot^2 \theta = \csc^2 \theta$$

Addition Formulas

$$\cos(A + B) = \cos A \cos B - \sin A \sin B$$

$$\sin(A + B) = \sin A \cos B + \cos A \sin B$$

Double-Angle Formulas

$$\cos 2\theta = \cos^2 \theta - \sin^2 \theta$$

$$\sin 2\theta = 2 \sin \theta \cos \theta$$

Half-Angle Formulas

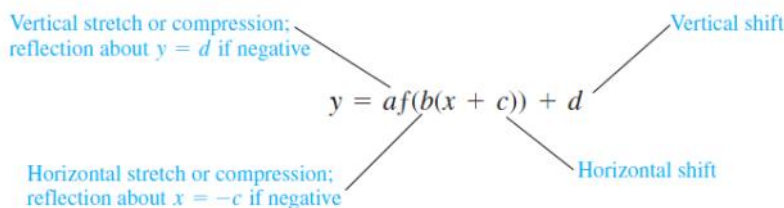
$$\cos^2 \theta = \frac{1 + \cos 2\theta}{2}$$

$$\sin^2 \theta = \frac{1 - \cos 2\theta}{2}$$

$$c^2 = a^2 + b^2 - 2ab \cos \theta.$$

Transformations of Trigonometric Graphs

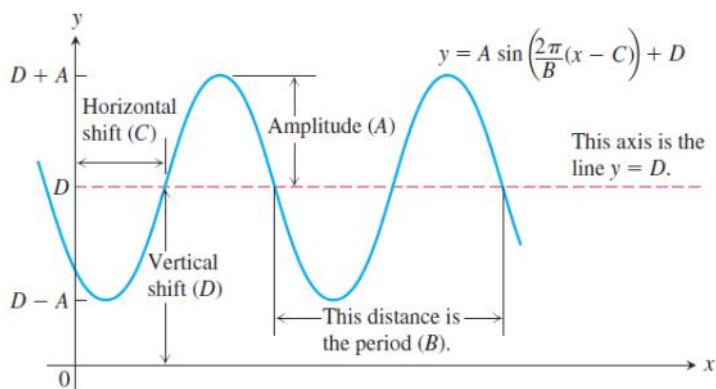
The rules for shifting, stretching, compressing, and reflecting the graph of a function summarized in the following diagram apply to the trigonometric functions we have discussed in this section.



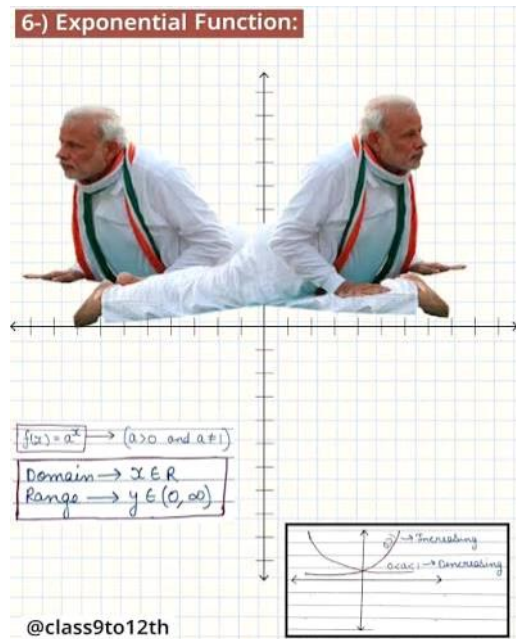
The transformation rules applied to the sine function give the **general sine function** or **sinusoid** formula

$$f(x) = A \sin\left(\frac{2\pi}{B}(x - C)\right) + D,$$

where $|A|$ is the *amplitude*, $|B|$ is the *period*, C is the *horizontal shift*, and D is the *vertical shift*. A graphical interpretation of the various terms is given below.



6-) Exponential Function:



Exponential Behavior

When a positive quantity P doubles, it increases by a factor of 2 and the quantity becomes $2P$. If it doubles again, it becomes $2(2P) = 2^2P$, and a third doubling gives $2(2^2P) = 2^3P$. Continuing to double in this fashion leads us to consider the function $f(x) = 2^x$. We call this an *exponential* function because the variable x appears in the exponent of 2^x . Functions such as $g(x) = 10^x$ and $h(x) = (1/2)^x$ are other examples of exponential functions. In general, if $a \neq 1$ is a positive constant, the function

$$f(x) = a^x, \quad a > 0$$

is the **exponential function with base a** .

EXAMPLE 1 In 2014, \$100 is invested in a savings account, where it grows by accruing interest that is compounded annually (once a year) at an interest rate of 5.5%. Assuming no additional funds are deposited to the account and no money is withdrawn, give a formula for a function describing the amount A in the account after x years have elapsed.

TABLE 1.5 Savings account growth

Year	Amount (dollars)	Increase (dollars)
2014	100	
2015	$100(1.055) = 105.50$	5.50
2016	$100(1.055)^2 = 111.30$	5.80
2017	$100(1.055)^3 = 117.42$	6.12
2018	$100(1.055)^4 = 123.88$	6.46

Rules for Exponents

If $a > 0$ and $b > 0$, the following rules hold true for all real numbers x and y .

1. $a^x \cdot a^y = a^{x+y}$
2. $\frac{a^x}{a^y} = a^{x-y}$
3. $(a^x)^y = (a^y)^x = a^{xy}$
4. $a^x \cdot b^x = (ab)^x$
5. $\frac{a^x}{b^x} = \left(\frac{a}{b}\right)^x$

EXAMPLE 2 We illustrate using the rules for exponents to simplify numerical expressions.

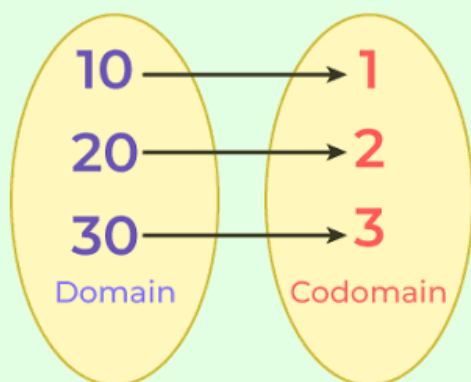
1. $3^{1.1} \cdot 3^{0.7} = 3^{1.1+0.7} = 3^{1.8}$ Rule 1
2. $\frac{(\sqrt{10})^3}{\sqrt{10}} = (\sqrt{10})^{3-1} = (\sqrt{10})^2 = 10$ Rule 2
3. $(5^{\sqrt{2}})^{\sqrt{2}} = 5^{\sqrt{2} \cdot \sqrt{2}} = 5^2 = 25$ Rule 3
4. $7^\pi \cdot 8^\pi = (56)^\pi$ Rule 4
5. $\left(\frac{4}{9}\right)^{1/2} = \frac{4^{1/2}}{9^{1/2}} = \frac{2}{3}$ Rule 5



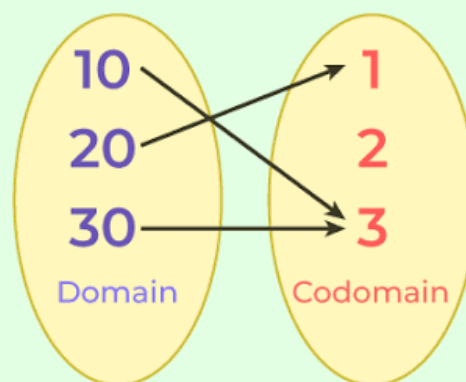
One-to-One Functions

DEFINITION A function $f(x)$ is **one-to-one** on a domain D if $f(x_1) \neq f(x_2)$ whenever $x_1 \neq x_2$ in D .

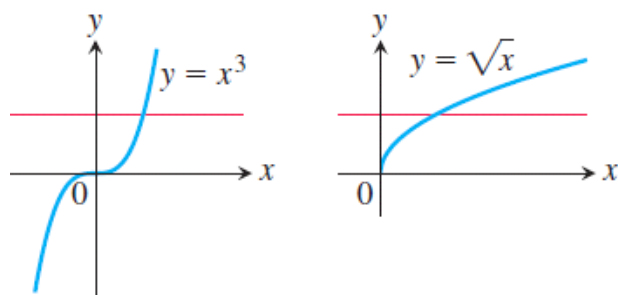
Injective Function



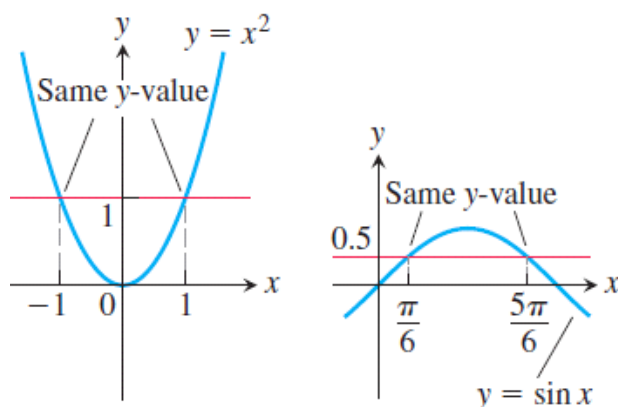
Injective



Not Injective



(a) One-to-one: Graph meets each horizontal line at most once.



(b) Not one-to-one: Graph meets one or more horizontal lines more than once.

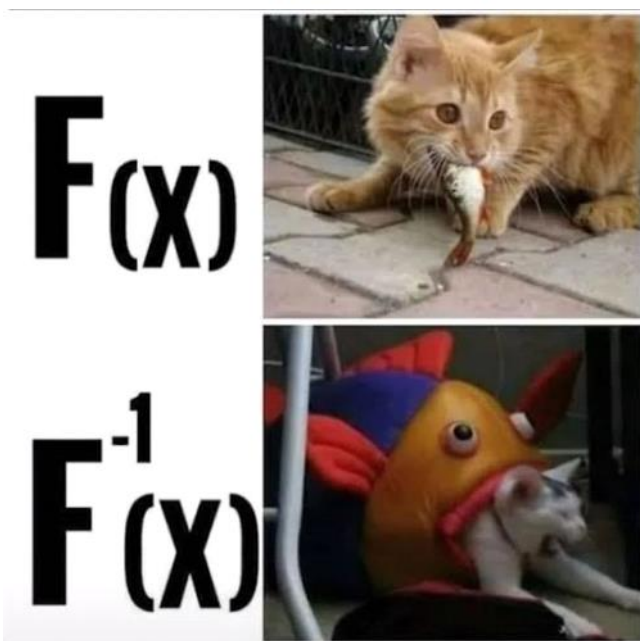
The Horizontal Line Test for One-to-One Functions

A function $y = f(x)$ is one-to-one if and only if its graph intersects each horizontal line at most once.

DEFINITION Suppose that f is a one-to-one function on a domain D with range R . The **inverse function** f^{-1} is defined by

$$f^{-1}(b) = a \quad \text{if} \quad f(a) = b.$$

The domain of f^{-1} is R and the range of f^{-1} is D .



Inverse Functions 🤔💀

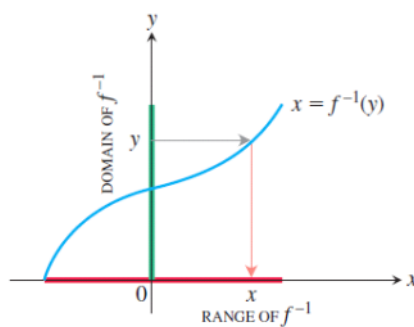
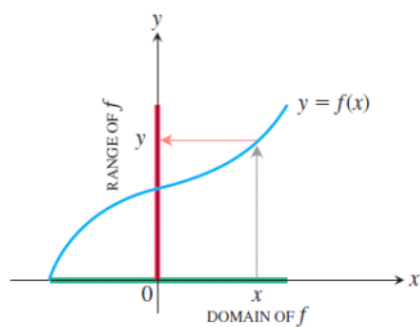


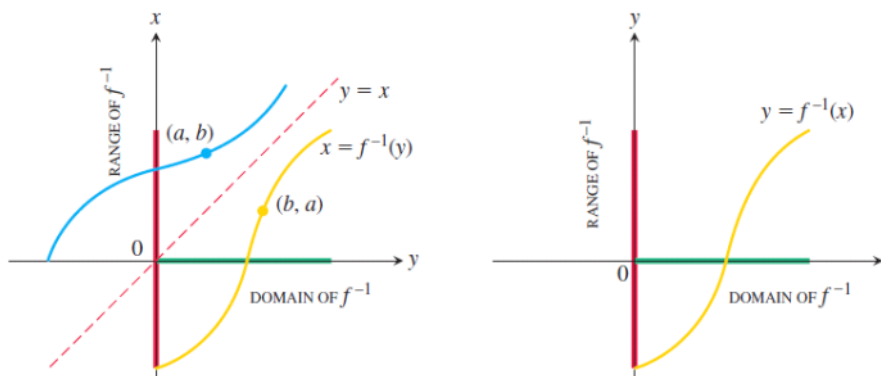
EXAMPLE 2 Suppose a one-to-one function $y = f(x)$ is given by a table of values

x	1	2	3	4	5	6	7	8
$f(x)$	3	4.5	7	10.5	15	20.5	27	34.5

A table for the values of $x = f^{-1}(y)$ can then be obtained by simply interchanging the values in the columns (or rows) of the table for f :

y	3	4.5	7	10.5	15	20.5	27	34.5
$f^{-1}(y)$	1	2	3	4	5	6	7	8





EXAMPLE 3 Find the inverse of $y = \frac{1}{2}x + 1$, expressed as a function of x .

Solution

1. Solve for x in terms of y : $y = \frac{1}{2}x + 1$ The graph is a straight line satisfying the horizontal line test (Fig. 1.60).

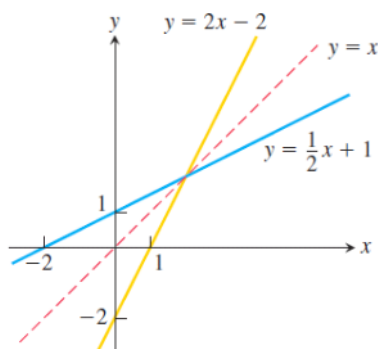
$$\begin{aligned} 2y &= x + 2 \\ x &= 2y - 2. \end{aligned}$$

2. Interchange x and y : $y = 2x - 2$.

The inverse of the function $f(x) = (1/2)x + 1$ is the function $f^{-1}(x) = 2x - 2$. (See Figure 1.60.) To check, we verify that both composites give the identity function:

$$f^{-1}(f(x)) = 2\left(\frac{1}{2}x + 1\right) - 2 = x + 2 - 2 = x$$

$$f(f^{-1}(x)) = \frac{1}{2}(2x - 2) + 1 = x - 1 + 1 = x.$$



EXAMPLE 4 Find the inverse of the function $y = x^2, x \geq 0$, expressed as a function of x .

Solution For $x \geq 0$, the graph satisfies the horizontal line test, so the function is one-to-one and has an inverse. To find the inverse, we first solve for x in terms of y :

$$y = x^2$$

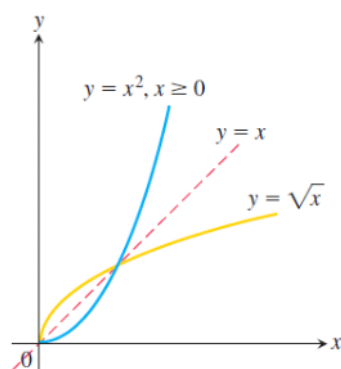
$$\sqrt{y} = \sqrt{x^2} = |x| = x \quad |x| = x \text{ because } x \geq 0$$

We then interchange x and y , obtaining

$$y = \sqrt{x}.$$

The inverse of the function $y = x^2, x \geq 0$, is the function $y = \sqrt{x}$ (Figure 1.61).

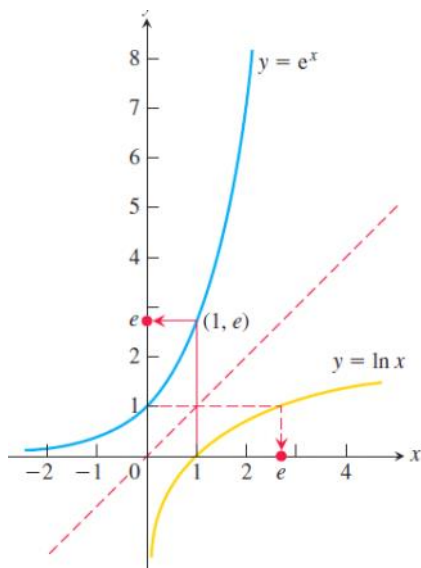
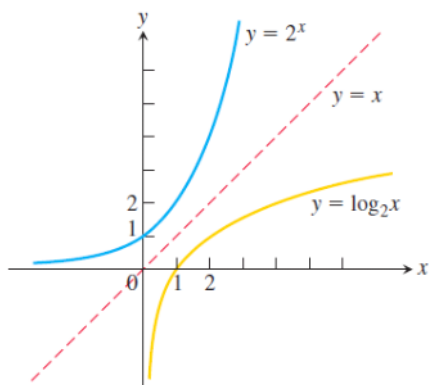
Notice that the function $y = x^2, x \geq 0$, with domain *restricted* to the nonnegative real numbers, *is* one-to-one (Figure 1.61) and has an inverse. On the other hand, the function $y = x^2$, with no domain restrictions, *is not* one-to-one (Figure 1.58b) and therefore has no inverse. ■



Logarithmic Functions

If a is any positive real number other than 1, the base a exponential function $f(x) = a^x$ is one-to-one. It therefore has an inverse. Its inverse is called the *logarithm function with base a* .

DEFINITION The **logarithm function with base a** , $y = \log_a x$, is the inverse of the base a exponential function $y = a^x$ ($a > 0, a \neq 1$).



$\log_e x$ is written as $\ln x$.

$\log_{10} x$ is written as $\log x$.

The function $y = \ln x$ is called the **natural logarithm function**, and $y = \log x$ is often called the **common logarithm function**. For the natural logarithm,

$$\ln x = y \Leftrightarrow e^y = x.$$

In particular, if we set $x = e$, we obtain

$$\ln e = 1$$

because $e^1 = e$.

THEOREM 1—Algebraic Properties of the Natural Logarithm For any numbers $b > 0$ and $x > 0$, the natural logarithm satisfies the following rules:

1. *Product Rule:* $\ln bx = \ln b + \ln x$
2. *Quotient Rule:* $\ln \frac{b}{x} = \ln b - \ln x$
3. *Reciprocal Rule:* $\ln \frac{1}{x} = -\ln x$ Rule 2 with $b = 1$
4. *Power Rule:* $\ln x^r = r \ln x$

EXAMPLE 5 Here we use the properties in Theorem 1 to simplify three expressions.

- (a) $\ln 4 + \ln \sin x = \ln(4 \sin x)$ Product Rule
- (b) $\ln \frac{x+1}{2x-3} = \ln(x+1) - \ln(2x-3)$ Quotient Rule
- (c) $\ln \frac{1}{8} = -\ln 8$ Reciprocal Rule
 $= -\ln 2^3 = -3 \ln 2$ Power Rule ■

Because a^x and $\log_a x$ are inverses, composing them in either order gives the identity function.

Inverse Properties for a^x and $\log_a x$

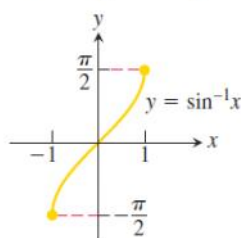
1. Base a : $a^{\log_a x} = x$, $\log_a a^x = x$, $a > 0, a \neq 1, x > 0$
2. Base e : $e^{\ln x} = x$, $\ln e^x = x$, $x > 0$

EXAMPLE 6 If \$1000 is invested in an account that earns 5.25% interest compounded annually, how long will it take the account to reach \$2500?

Inverse Trigonometric Functions

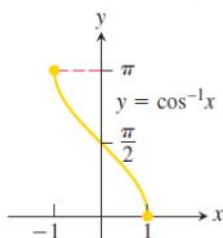
$$\begin{array}{ll} y = \sin^{-1} x & \text{or} \quad y = \arcsin x \\ y = \cos^{-1} x & \text{or} \quad y = \arccos x \\ y = \tan^{-1} x & \text{or} \quad y = \arctan x \\ y = \cot^{-1} x & \text{or} \quad y = \operatorname{arccot} x \\ y = \sec^{-1} x & \text{or} \quad y = \operatorname{arcsec} x \\ y = \csc^{-1} x & \text{or} \quad y = \operatorname{arccsc} x \end{array}$$

Domain: $-1 \leq x \leq 1$
Range: $-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$



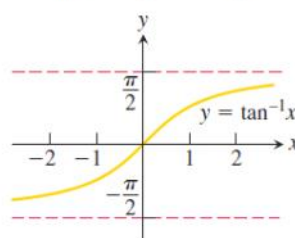
(a)

Domain: $-1 \leq x \leq 1$
Range: $0 \leq y \leq \pi$



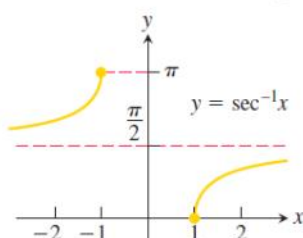
(b)

Domain: $-\infty < x < \infty$
Range: $-\frac{\pi}{2} < y < \frac{\pi}{2}$



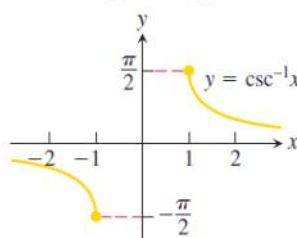
(c)

Domain: $x \leq -1$ or $x \geq 1$
Range: $0 \leq y \leq \pi, y \neq \frac{\pi}{2}$



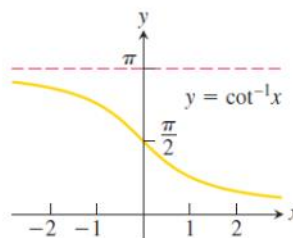
(d)

Domain: $x \leq -1$ or $x \geq 1$
Range: $-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}, y \neq 0$



(e)

Domain: $-\infty < x < \infty$
Range: $0 < y < \pi$



(f)

